

Week 7 Week 7 Add one challenge — paper completion route (no device) · paper route

Paper route · GROW Computing · 40 minutes

Name: _____ Date: _____

Week 7. Add one challenge. Paper route.

I can add one new feature that changes how the game is played.

This route reaches the same goal as the screen route. You do it with paper.

You will design a moving Hazard. You will write its blocks. You will write the contact rule on Player. You will test it on the printed maze.

Point, say, draw or write. An adult may write your exact words.

Week 6 note. On the screen route you open your own W06_Win.sb3. On this route no .sb3 file was ever made. Open your dated Week 6 grid pages from your folder instead. Your adult will not ask you for a file you never made.

What you need. The printed maze page. A green counter for Player. A red diamond counter for Hazard. The Hazard clock strip. The block grids. The design sheet. A pencil. No device.

The touching rule on paper. Two counters are touching if they sit on the same crossing. They are also touching if they sit on crossings next to each other, corner to corner as well. Two crossings apart is not touching.

One small square on the maze is 10 units. Five small squares are 50 units.

The clock. One clock step is half a second. In one clock step the Hazard moves five small squares.

The play rule. Move the Player counter five small squares. Then move the Hazard counter one clock step. Say touching or not touching after each turn.

Saving. There is no .sb3 file on this route. Write Week 7 Add one challenge at the top of every page. Write your name. Write the date. Put the pages in your folder. Your adult will initial the folder.

Point, say, draw or write. Someone may record your exact words.

On the screen route this week is saved as W07_Challenge.sb3. On the paper route no file is made. Your pages are the record: write your name and the date on every page and put them in your folder. An adult initials the folder. Nobody should ask you for W07_Challenge.sb3 if you took the paper route.

Write your code

Hazard · Grid 1. The new feature. A moving obstacle that repeats its journey on every run. Write one block on each row. Both glide blocks go inside forever. Show inside by moving the writing one step to the right.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked

Blocks you may use this week:

- go to x: _ y: _
- glide _ secs to x: _ y: _
- forever
- if _ then
- change y by _

Player · Grid 2. The new contact rule. This is a second, separate stack. It sits beside the wall rule. It is not inside it. The consequence belongs to Player because Player is the sprite that must move.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked

Blocks you may use this week:

- forever
- if _ then
- touching [Hazard]?
- go to x: _ y: _
- wait until _
- say _

Player (carried forward from Week 6 — must still work after the change) · Grid 3. Reset to the safe start on the flag. Copy this from your Week 6 pages. Check it is still there after your change.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked

Blocks you may use this week:

- go to x: _ y: _
- change x by _
- change y by _

Player (carried forward from Week 6 — must still work after the change) · Grid 4. The four arrow controls. Four separate hat scripts. The grid is printed four times on one page, one copy per key. Each copy has three rows: one block row and two spare rows. Twelve rows in total.

Write one block on each line. Start with the hat block, which is done for you.

when [right arrow] key pressed — then the same grid again for [left arrow], [up arrow] and [down arrow]

Blocks you may use this week:

- change x by _
- change y by _
- go to x: _ y: _
- when [right arrow] key pressed
- when [left arrow] key pressed

- when [up arrow] key pressed
- when [down arrow] key pressed

Player (carried forward from Week 6 — must still work after the change) · Grid 5. The wall rule. Write it out again. It is built the same way as your new Hazard rule. That is why the new rule is its own stack and not an edit of this one.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked

Blocks you may use this week:

- forever
- if _ then
- touching [Maze]?
- go to x: _ y: _
- wait until _

Player (carried forward from Week 6 — must still work after the change) · Grid 6. The win response. The empty say block goes first. It clears the old bubble when the game restarts.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked

Blocks you may use this week:

- say _
- go to x: _ y: _
- wait until _
- touching [Finish]?
- forever
- if _ then

Hazard — bug card (read only) plus a repair grid headed Player · Grid 7. The bug hunt, on paper. The top half is a card headed Hazard. The faulty script is already printed on it. Do not change the printed card. Read it. Say what it does wrong. The bottom half is an empty grid headed Player. Write the repaired contact rule there. Then draw one line through the Hazard card to show that script is removed.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked (printed on the Hazard card) — then the empty repair grid uses the same hat block, headed Player	

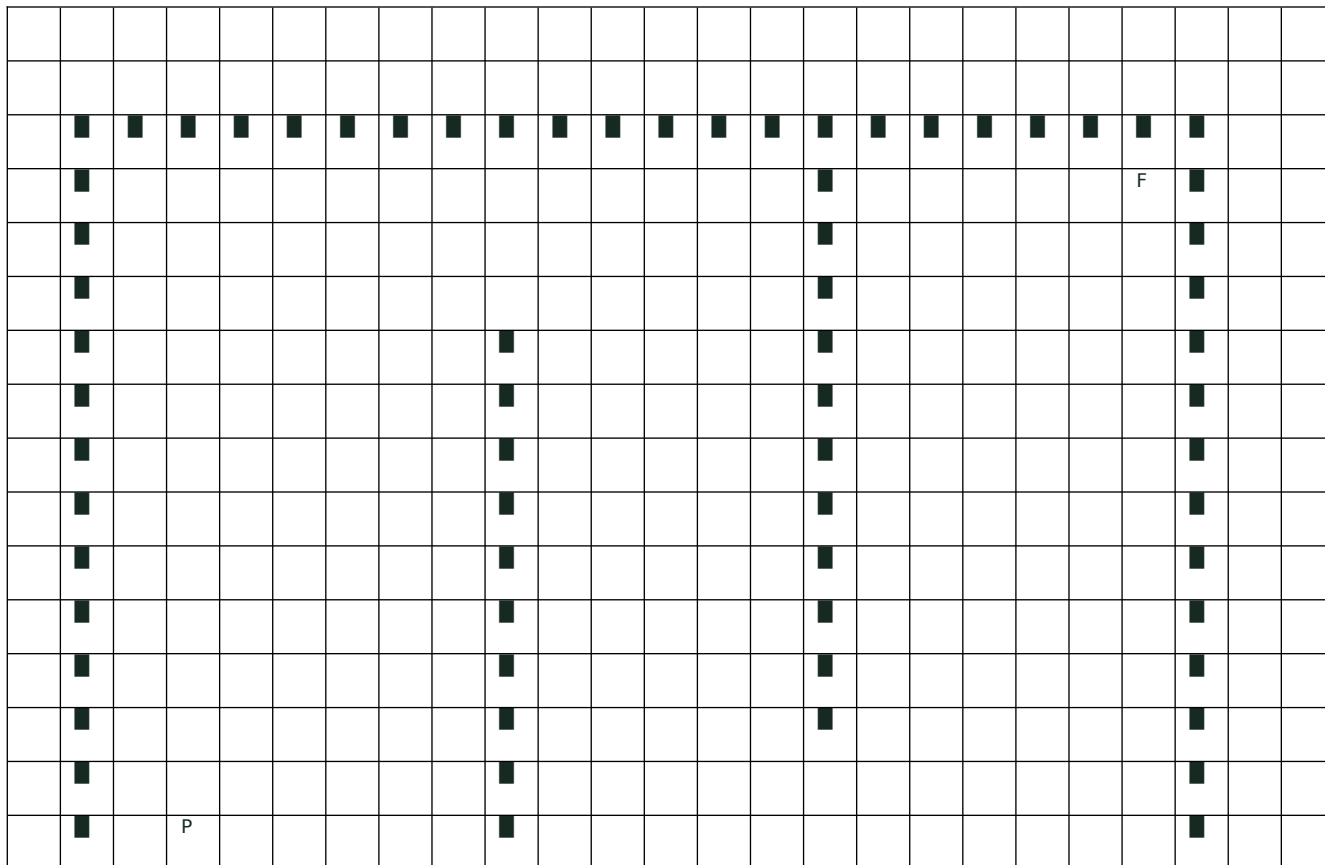
Blocks you may use this week:

- forever
- if _ then
- touching [Hazard]?
- touching [Player]?
- go to x: _ y: _

The maze

The printed maze is the real game at its real size. The stage is 480 units across and 360 units down. x runs from -240 on the left to 240 on the right. y runs from -180 at the bottom to 180 at the top. One small square is 10 units. Numbers are printed along the edges every 20 units. Six walls are drawn at their real coordinates: the bottom wall, the top wall, the left wall, the right wall, a tall wall at x -70 to -50 that rises from the bottom and stops at y 80, and a tall wall at x 50 to 70 that hangs from the top and stops at y -80. Put a counter on a crossing of the grid lines. Read its x from the bottom edge and its y from the side edge. STEP 1: write your name and the date at the top. STEP 2: put the green Player counter on the safe start and write the two numbers beside it. STEP 3: find Finish and write its two numbers beside it. STEP 4: draw the Hazard's line. It is a straight line at x 0 from y -100 up to y 100. Put a small arrow at each end. STEP 5: put the red diamond Hazard counter at the bottom end of the line. That is where the Hazard goes on the flag. STEP 6: shade the low band. It is the part of the Hazard's line from y -100 up to y -50. It is five small squares tall. Write low band beside it. STEP 7: draw one route from the safe start to Finish that a player can use. Put a dot at every turn. Write the two numbers at every dot. STEP 8: look at where your route passes the Hazard's line. Write how many small squares away you are. Use the touching rule and write touching or not touching. STEP 9: play the route. Move the Player counter five small squares. Then move the Hazard counter one clock step. Say touching or not touching. Keep going until Player reaches Finish. Your adult watches and counts your turns. STEP 10: play it again down the middle of the maze at x 0 instead. Say what is different.

The maze below is the real one, at its real size. One square is 20 steps. Player starts at (-180, -120). Finish is at (180, 120). Move a counter square by square and write down where it lands.



[illegible]

Predict, trace, compare

Trace 1. The Hazard script. One row for each block. Write where the Hazard is when the block has finished.

Row	Block I run	Time now (seconds)	Hazard x	Hazard y
1	when green flag clicked	0		
2	go to x: 0 y: -100	0	0	-100
3	glide 2 secs to x: 0 y: 100			
4	glide 2 secs to x: 0 y: -100			
5	forever repeats. glide 2 secs to x: 0 y: 100			
6	glide 2 secs to x: 0 y: -100			

Trace 2. The Hazard clock. One clock step is half a second. The Hazard moves five small squares each step.

Clock step	Time (seconds)	Hazard x	Hazard y	In the low band?
0	0.0	0	-100	yes
1	0.5			
2	1.0			
3	1.5			
4	2.0			
5	2.5			
6	3.0			
7	3.5			
8	4.0			

Trace 3. The Player arrow script. One row for each block. Write where Player is when the block has finished.

Row	Block I run	Player x	Player y
1	when green flag clicked		
2	go to x: -180 y: -	-180	-120

	120		
3	when [up arrow] key pressed, change y by 10		
4	when [up arrow] key pressed, change y by 10		
5	when [right arrow] key pressed, change x by 10		
6	when [right arrow] key pressed, change x by 10		
7	when [down arrow] key pressed, change y by -10		

Predict and trace. The four shared setups. Write your prediction first. Then move the counters. Then write what you traced.

Setup	I predict	I traced	Same?
a. Green flag. Player counter on (0, 0). Hazard counter on (0, -100). Advance the clock 2 steps.			
b. Green flag. Player counter on (30, 0). Advance the clock 2 steps.			
c. Green flag. Advance the clock 4 steps. Then advance 4 more steps.			
d. Green flag after a win. Trace the win script. Then put Player on the wall square (-60, 0). Then put Player on Finish (180, 120).			

Before and after. The same setup on both games. This shows what my new feature changed.

Setup	Week 6 game, no Hazard	Week 7 game, Hazard added
Green flag. Player counter on (0, 0). Advance the clock 2 steps.		
Green flag. Player counter on (30, 0). Advance the clock 2 steps.		
What the player must now do that they did not do before		

Compare two settings. Use your own traced results, not a feeling. The low band is five small squares tall.

Setting	Small squares the Hazard moves in one second	Seconds in the low band on one visit	Seconds the low band is clear between visits	Could I cross? What my test showed
glide 2 secs to each end				
glide 4 secs to each end				

Trace 4. The bug card. The rule is on the wrong sprite. Trace it and say what goes wrong.

Row	What happens	Player x	Player y	Hazard x	Hazard y
1	Green flag. Player counter on (0, 0). Hazard counter on (0, -100).	0	0	0	-100
2	Advance the clock 2 steps.				
3	The if block is on Hazard. It says go to x: -180 y: -				

	120.				
4	Advance the clock 1 more step. The glide carries on.				

This week only

DESIGN SHEET. My moving Hazard.

My sprite name: Hazard. I wrote this name myself.

My drawing of the Hazard. Draw it in the box. The model Hazard is a small red diamond, 20 units across. That is two small squares.

MY HAZARD JOURNEY.

On the flag the Hazard goes to x 0, y -100.

Then it glides 2 seconds to x 0, y 100.

Then it glides 2 seconds back to x 0, y -100.

Both glide blocks sit inside forever. That is why the journey repeats on every run.

The whole journey is 200 units long. That is 20 small squares.

THE CONTACT RULE BELONGS TO PLAYER.

Player checks touching [Hazard]?

Player goes to its safe start, x -180, y -120.

Write it again in your own words: the check goes on Player. The go to goes on Player. Player is the sprite that must move.

If you put the rule on Hazard, the Hazard moves and the player stays where it is. Then the feature does nothing to the player.

CHECK MY JOURNEY IS FAIR.

Is the journey clear of the safe start? The start is at x -180. The journey is at x 0. Yes, 18 small squares away.

Is the journey clear of Finish? Finish is at x 180, y 120. Yes.

Can a player still reach Finish? Yes. Go down the middle at x 30. That is 3 small squares from the line. The touching rule says not touching.

If every route is blocked, make the Hazard smaller or move its two ends.

IS SLOWER FAIRER? TEST IT, DO NOT GUESS.

Setting 1: glide 2 secs to each end. The Hazard moves 10 small squares in one second.

Setting 2: glide 4 secs to each end. The Hazard moves 5 small squares in one second.

The low band is 5 small squares tall. It is the part of the journey nearest the bottom crossing.

On setting 1 the Hazard is in the low band for 1 second on each visit.

On setting 2 the Hazard is in the low band for 2 seconds on each visit.

So the slow Hazard sits on the crossing for twice as long.

Your own crossing takes the same time on both settings. You did not get slower.

The wait between visits is longer too: 3 seconds on setting 1, 6 seconds on setting 2.

So slower is not automatically fairer. It gives a longer clear gap, and a longer block.

Write the setting you chose: _____

Write the test result that made you choose it: _____

Write what a player said about it: _____

MY OWN MAZE. If you are not using the printed maze, choose two clear end points for the journey. Keep both ends away from the safe start. Keep the whole Hazard clear of Finish. Write your two end points here: (____ , ____) and (____ , ____).

MY SAFE START: x _____ y _____

MY IDEA AND ITS EFFECT. Look again at your Week 2 mission idea. Which feature can you build today? Agree a change with your adult and say why.

What I said: _____

What the adult said back, and what changed: _____

My completion record

What this week shows

AQA 71638 Programming with Scratch (unit 6), Level One. Outcome 5: added a feature that changes gameplay. The pack's teacher guide states the evidence: "Observe the pupil creating the feature and explaining what the player must now do. Record a before-and-after test. Decoration alone does not meet the outcome." Evidence for the outcome is a summary sheet completed by the teacher from individual observation. The centre's AQA summary sheet remains the official record and is completed by staff. The statements below are a local scaffold only. The route the pupil took is recorded in the teacher guide, never on the AQA summary sheet.

- ☐ I created a new sprite and named it myself. On paper: I drew the Hazard on the design sheet, I wrote the name Hazard at the top of its own block grid, and I cut out and placed its counter.
- ☐ My feature moves or responds on every run, not only once. On paper: I wrote forever with both glide blocks inside it, and I ran the clock strip twice all the way round to show the journey comes back to (0, -100) and goes again.
- ☐ Contact with my feature has a consequence the player can see. On paper: I traced a contact setup, I moved the Player counter back to (-180, -120), and I said out loud why it moved.
- ☐ The four controls, the wall rule and the win still work after my change. On paper: I wrote out all four arrow scripts, the wall rule and the win response, and I traced one arrow run, one wall square and one win.
- ☐ Extra, from the design sheet: I can say what the player must now do that they did not do before.
- ☐ Extra, from the compare table: I chose my glide setting from a test result I wrote down, not from a feeling.

Adult record

- ☐ Watch the pupil write Grid 1 and Grid 2. The pupil must place both glide blocks inside forever, and must place the contact check on Player and not on Hazard. Ask which sprite the go to block belongs to. The answer must be Player.
- ☐ Watch the pupil name the Hazard. Naming and drawing must be the pupil's own, not copied from a printed model. Record which parts the pupil produced today.
- ☐ Ask the discriminating question: what must the player now do that they did not do before? A pupil who answers only "it looks different" has described decoration, and outcome 5 is not met. A pupil who says they must watch the Hazard, wait, or go round the side has met it.
- ☐ Record the before-and-after test from the before and after table. The same setup, two different results, is the required record.
- ☐ POSE AT LEAST ONE UNSEEN SETUP. The four shared setups have their expected results printed in this guide, so a pupil could copy them. An unseen setup shows the pupil is applying the rule. Use one of these three. U1: green flag, Player counter on (0, 110), advance the clock 4 steps. U2: green flag, Player counter on (30, -110),

advance the clock 8 steps. U3: green flag, Player counter on (0, -110), do not move the clock at all.

☐ Accept pointing at the counter, saying the answer, drawing the move, or writing it. Any of the four is a full response.

☐ Record the date and the support used against each pupil statement.

☐ Record in the teacher guide that this pupil took the paper route. Do not write the route on the AQA summary sheet and do not alter that sheet.

☐ There is no W07_Challenge.sb3 on this route. Initial the pupil's dated Week 7 pages in the folder. Week 8's paper route opens those pages.

Adult name: _____ Date: _____

Route taken: paper · Recorded in the teacher guide, not on the AQA form.